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INTEREST RATES IN THE NINETEEN-FIFTIES

Thomas J. Sargent *

ECONOMISTS have recently begun to devote an increasing amount of attention to the relationships among interest rates on various instruments. Rates on instruments which differ with respect to maturity alone, all other features supposedly being held constant, have in particular received a great deal of attention since the publication of Meiselman's [7] work in 1962. While the term structure has certainly received the most concern, important contributions have been made to the broader problem of studying relationships among rates on bonds which differ with respect to features other than maturity.¹ An especially important aspect of this broader area of research lies in examining the relationships between rates on government and corporate bonds. Knowledge of the characteristics of these relationships is important for an understanding of the paths through which monetary policy affects yields on corporate bonds, and hence, perhaps, expenditures on investment.

This paper presents the results of an examination of the relationships among several interest rates on government and corporate bonds for the period January 1951 through December 1960. Monthly data are used, and series for the rates on three-month treasury bills, one, two, three, four, five, ten, and twenty-year government bonds, commercial paper, and Moody's Aaa's and Baa's are studied.² This list represents an array of instruments ranging over a broad maturity spectrum and featuring

various levels of quality. Tools of spectral and cross-spectral analysis are used to study the behavior of the series and the relationships among the series at various important components of oscillation.³ In addition to calculating the standard statistics associated with the spectrum and cross-spectrum, the relatively new tool of complex demodulation is employed to study the seasonal behavior of selected rates.⁴

I Spectral Results

The spectral density of a time series gives the variance in the series occurring at each of several frequency bands. The presence of a peak at some frequency indicates that an important component at this frequency band is present in the series. To investigate the periodicities present, spectral densities were estimated for each of the eleven series studied. To diminish the distortions produced by the trend characterizing interest rates over this period, the spectral density of the first differences, rather than the levels, of each series was calculated. In each case, thirty was the maximal lag employed in the calculations. Space limitations prevent us from showing all eleven estimated densities, but the central characteristics of the results can be illustrated with figures 1(a-d) which show the estimated spectral densities for first differences in the treasury bill, five and twenty-year government, and Baa rates:

a) The spectral density of the bill rate clearly shows the presence of a seasonal component. Peaks occur at twelve, six, four, three, and two months, which correspond to the seasonal component and its first four harmonics. In addition, the small peak at thirty months re-

* The author thanks Peter Frost and a referee for helpful comments.

¹ For example, see Okun [8] and Conard [11].

² The data were taken from two sources. From the *Federal Reserve Bulletin*, monthly averages of long term government, Aaa, Baa, treasury bill, and commercial paper rates were taken. From the Salomon Brothers and Hutzler pamphlet *An Analytical Record of Yields and Yield Spreads, Part I*, rates on three month bills and one, two three, four, five, ten, and twenty-year governments were obtained. The rates were mid-month observations from 1951-1958, and first-of-month for 1959-1960. This change in the recording convention induces a mild distortion in the data, but the consequences for the results given here are undoubtedly very small. It was necessary, however, to avoid comparing the Federal Reserve series with the Salomon Brothers data, since the differences in their construction would seriously distort the timing of relationships among series.

³ Granger and Hatanaka [4] contains a treatment of these techniques.

⁴ The spectra and cross-spectra were obtained by the standard Blackman-Tukey covariance-cosine transformation procedures. The calculations were performed by using the ALGOL procedure SPECTRUM which was written by the author for the Carnegie Tech. G-21 computer. The calculation of the complex demodulates was performed using the ALGOL procedure COMPD which was also written for the Carnegie Tech. G-21.

veals the possible presence of a cyclical component.⁵

b) As the spectral densities of the five and twenty-year governments indicate, the strength of the seasonal declines with increases in maturity. This is shown by the diminished stature of the peaks occurring at twelve months and by the weaker peaks which occur near the harmonics of the seasonal. In addition, while small peaks exist at the cyclical frequency, there is a tendency for the higher frequency oscillations to become more important relative to the lower as the maturity increases.

c) The spectral density of the Baa index in important respects is similar to those of the government bonds of intermediate maturity. Peaks occur at thirty and twelve months. In the case of the higher frequencies, however, the Baa's spectral density has peaks at the harmonics of the cyclical frequency rather than at or near the harmonics of the seasonal, as do the governments.

In summary, the spectral densities of the eleven interest rates under study reveal some evidence that common periodicities are present in the series at the cyclical and seasonal frequencies. However, it is clear that the strength of these periodicities, particularly at the seasonal, varies a great deal among instruments and generally diminishes with increases in maturity. This does not necessarily indicate that the seasonal patterns in the rates on longer maturity bonds were any less regular than those on the shorter maturity instruments. It may simply reflect the smaller amplitude of fluctuation which characterizes rates on longer term bonds. These differences in amplitude, as well as other features of the relationships among series, can be studied with cross-spectral analysis, to which we now turn.

⁵ It should be noted that the use of first differences, which usually eliminates most of the power at the very long periodicities, can produce a spurious peak at some long periodicity provided that the original process has a spectrum whose power falls off fairly regularly with increasing frequency, which is the typical shape of spectra of economic variables. It is possible, therefore, that the peaks at thirty months and twelve months are spurious. It happens, however, that evidence for the presence of these periodicities exists in estimated spectra of the original series. All of the results reported in this paper remain substantially unaltered when the analysis is carried through without first differencing the series.

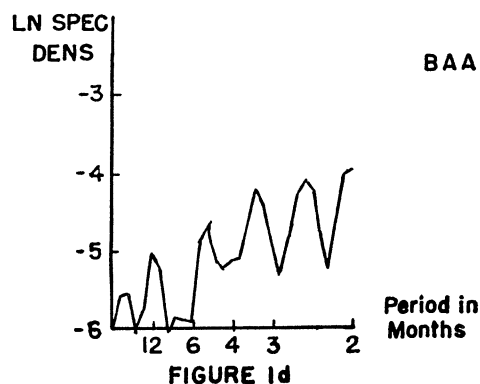
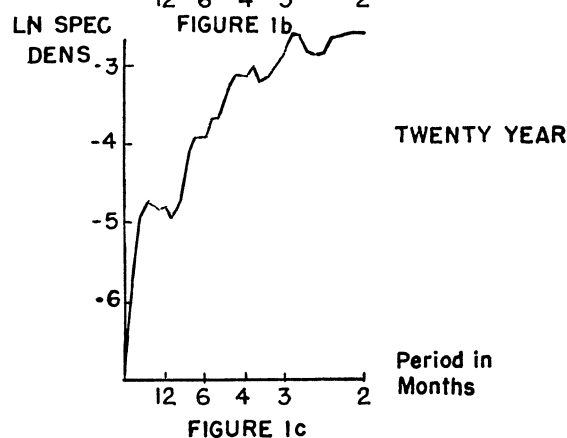
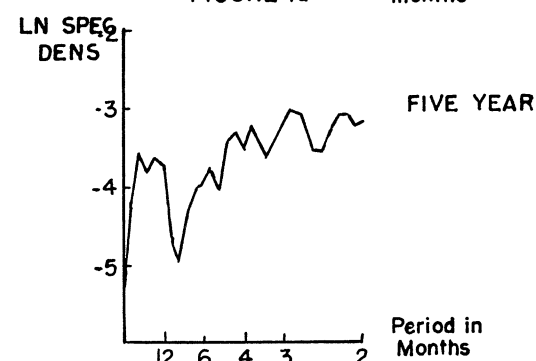
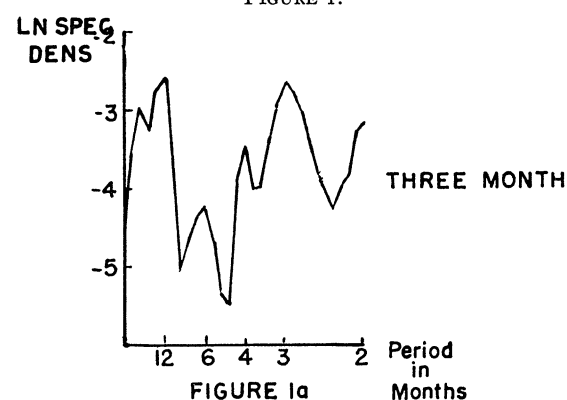
II Cross-Spectral Results

It was pointed out above that the spectral density gives the variance of a series which occurs over each of a number of frequency bands. Similarly, the empirical cross-spectrum provides an estimate of the covariance between two series occurring over each frequency band. Unlike the spectral density, however, the value of the cross-spectrum at each frequency is a complex number. The general practice has therefore been to characterize the cross-spectrum with one or more of three statistics which are themselves real numbers. These are the coherence, phase shift, and gain. The coherence is similar in concept to the R^2 of correlation analysis. The R^2 statistic is simply the covariance between two series squared divided by the product of their variances. A measure of covariance between two series at each frequency is provided by the amplitude of the cross-spectrum. The amplitude can be thought of as the sum of the covariances in phase and in quadrature. Dividing the squared amplitude by the product of the estimated spectral densities at a given frequency provides a measure of the proportion of the variance in one series at this frequency explained by the component of similar frequency in the other series, disregarding any phase differences which might exist. This statistic is the coefficient of coherence. The analogy to the R^2 statistic is obvious, and like the R^2 statistic, the coherence is bounded by zero and one.

The phase shift statistic provides an estimate of the average lead or lag of one series over another at each frequency band under study. It is equal to the argument of the cross-spectrum at each frequency.

The gain statistic records the scalar by which the amplitude of one series at some frequency must be multiplied to produce a component of the amplitude at which this frequency appears in another series. It is analogous to the simple regression coefficient, which is simply the ratio of a covariance to the variance of the independent variable. Similarly, the gain at any frequency is given by the ratio of the amplitude of the cross-spectrum to the value of the spectral density of the "independent" variable at this frequency.

FIGURE 1.



This set of three statistics provides us with the tools to study the strength of association and time structure of relationships among series at various frequencies. In addition, with the gain, differences in amplitudes among series at various components of oscillation can be determined. Accordingly, cross-spectra were calculated for each of the longer government rates with the three-month bill rate. Once again, first differences were used and thirty was the length of the maximal lag. The results for the low frequencies are reported in table 1.

The coherences, which are almost all very high, reveal some important patterns. The strength of association between the series at any given frequency generally declines with increases in the maturity difference between series. In addition, the coherence between any two series declines as the period of oscillation diminishes. Thus, for example, rates were less highly correlated at the seasonal than at the thirty-month component. Nevertheless, what stands out in these figures is the very high association among first differences of rates which existed over a large part of the government yield curve and at most of the important components of oscillation.

The gain statistic recorded is the gain of the treasury bill rate over the longer rate in each instance, i.e., the coefficients β_{jn} in the "regressions" $\Delta S_t(w_j) = \beta_{jn} \Delta R_{nt}(w_j)$ are recorded where S_t is the bill rate and R_{nt} the rate on n year securities at time t . Not surprisingly the figures reveal that the amplitude of fluctuations at each frequency generally diminishes with increases in maturity. More interesting, perhaps, are the considerable differences which characterize the gain statistics of the bill rate against the long rate at the various frequencies. Consider the implications of these differences in the context of estimating the simple regression equation $\Delta S_t = \beta \Delta R_{nt} + \epsilon_t$ where ϵ_t is a random term with mean zero. The least squares estimate of β can be thought of as a certain weighted average of the regression or gain coefficients at each frequency, the weights depending upon the relative importance of the various frequencies. If the relative importance of these frequencies changes over time, then even if the gain at each frequency is unchanged, the simple regression coefficient will change as

TABLE 1. — FIRST DIFFERENCES (1951–1960)

Period	Coherence						
	3 mo 1 yr	3 mo 2 yr	3 mo 3 yr	3 mo 4 yr	3 mo 5 yr	3 mo 10 yr	3 mo 20 yr
60.0	.9974	1.0000	.9992	.9917	.9797	.9094	.8296
30.0	.9938	.9795	.9637	.9620	.9583	.9209	.8162
20.0	.9671	.9100	.8526	.5875	.8523	.8156	.6215
15.0	.9469	.8614	.8022	.7919	.7791	.7375	.5694
12.0	.9333	.8378	.7709	.7332	.6908	.5878	.3916
Phase Shift ^a							
60	-.0223	-.0424	-.0698	-.0065	-.0683	-.0326	-.0970
30	-.0221	-.0444	-.0751	-.0889	-.0996	-.1188	-.1419
20	.0424	.0492	.0196	-.0129	-.0370	-.1394	-.0519
15	-.0047	.0199	-.0508	-.1230	-.1664	-.3410	-.1803
12	-.1660	-.1882	-.3518	-.4836	-.5603	-.7978	-.7225
Gain ^b							
60	.9474	1.0797	1.1957	1.3316	1.4506	2.2641	2.8863
30	.9216	1.0141	1.1081	1.2249	1.3182	1.9818	2.4438
20	.9295	.9796	1.0098	1.1051	1.1574	1.5701	1.6907
15	1.1361	1.2261	1.2788	1.3528	1.3885	1.8775	2.1681
12	1.2753	1.4669	1.5100	1.5119	1.4868	1.7669	1.8983

^a Recorded in radians.^b Gain of bill rate with respect to long rate.

the time period is varied. The differences of the gain across frequencies therefore provides a possible explanation for the instability in the relationship between longs and shorts which Fand [2], among others, has found. The apparent instability of the relationship between rates may partly reflect the changing “mix” of oscillations rather than any change in the “regression” coefficients at each frequency.

The phase statistics reported in table 1 record the shift of the longer rate relative to the bill rate in each instance. Where a negative sign precedes the figure, the implication is that the longer rate leads the bill rate at this frequency; otherwise the bill rate leads. The predominance of negative signs suggests that the longer rates generally lead the bill rate.⁶ In particular, notice the pattern of leads across maturities which emerges at the thirty and twelve-month oscillations. At these fre-

quencies, the lead of longs over bills generally increases with the maturity of the long rate. This pattern is reminiscent of the one found by Macaulay [6] for the 1890–1913 period when a seasonal also existed in money market rates. Macaulay found that time rates led the shorter maturity call rates, and he interpreted this as evidence of successful forecasting within the framework of the expectations theory of the term structure. On the expectations hypothesis, the time rate actually had a forecast of the shorter rate impounded within it, and Macaulay reasoned that it should lead the call rate provided that expectations were reasonably accurate. Since the seasonal had persisted for a number of years, there was reason to expect it to be incorporated into operators’ expectations.

However, the similarity to Macaulay’s results is limited in important respects. While the lead of the one-year rate over the three-month rate at the seasonal would be expected according to Macaulay’s argument, there is no reason that this lead should increase with the maturity of the longer rate. In fact, once the maturity of the instrument is one year or longer Macaulay’s mechanism supplies no reason to expect the lead to increase as the maturity increases in steps of one year or some integer multiple of one year. Similarly, Macaulay’s

⁶ Since the coherence coefficient can be thought of as an inverse index of the variance of the phase statistic, it is clear that the significance of these phase differences depends on the coherence at the various frequencies. More precisely, the asymptotic variance of the phase at frequency w_j is approximately given by

$$\frac{m}{2n} \left[\frac{1}{\text{coh}(w_j)} - 1 \right]$$

where m is the maximal lag, n is the number of observations, and $\text{coh}(w_j)$ is the coherence at frequency w_j (see Hinich and Clay [5]). Similarly, the coherence is also an inverse index of the variance of the gain statistic.

argument provides no explanation for the tendency of the lead of longs over shorts to increase monotonically at the thirty-month cycle as the maturity of the long instrument extends to three years and beyond. It therefore appears that we must inquire elsewhere for the causes of this phenomenon.

A possible explanation for the pattern at the seasonal lies in the behavior of the average maturity of the publicly held government debt over these years. The average maturity is held to be a determinant of the spread between long and short rates in segmented markets and hedging theories of the term structure. These theories hold that, other things equal, the shorter the average maturity, the higher short rates must be relative to long rates in order to induce investors to hold the existing supplies of all maturities. The spread between long and short rates is thus supposed to be positively related to the average maturity of the publicly held debt.⁷ The relevance of this here is seen by noting that the phase differences existing at the seasonal component suggest that there is probably a significant seasonal component in the rate spread. The question then arises whether there is a seasonal pattern in the average maturity which might explain the seasonal in the rate spread. The answer is given in figure 2a which records the estimated spectral density of first differences in the average maturity of the publicly held government debt calculated from monthly data for the period January 1955 to December 1960.⁸ Twenty-four was the maximal lag. For the same period

⁷ Okun [8] discusses such a theory. It should be noted, however, that a severe identification problem characterizes these models. This occurs because suppliers of debt may vary the maturity structure of the debt as the general structure of yields rises and falls. To minimize the interest cost of the debt, suppliers increase the average maturity of the debt when long rates are low, and decrease the average maturity when long rates are high. Given the inverse relation between the spread between long and short rates and the level of long rates, such a mechanism produces a positive relationship between the average maturity and the rate spread. It then becomes difficult to determine whether the existence of a direct relationship between the average maturity and the rate spread arises from the impact of rate levels on the average maturity or from the adjustment of rate spreads in response to variations in average maturity which are necessary to induce investors to hold existing supplies willingly.

⁸ These data are available on a monthly basis only from January 1955 forward. The data are end-of-month observations.

and the same maximal lag, the spectral density of first differences of the spread between the ten-year government and the bill rate was calculated, and this is given in figure 2b. The figures confirm the presence of a seasonal in both series. For the average maturity, notice that the four and three-month frequencies contain the highest peaks. Figure 2c, which graphs the coherence between these two series, demonstrates that the association is fairly high at the seasonal components. In addition, the coherence generally has peaks at the seasonal frequencies.

To ascertain whether the seasonal components of the rate spread and average maturity were positively associated as implied by the segmented markets or hedging theory, the seasonal patterns of the two series were studied by means of the tools of complex demodulation.⁹ Use of the technique of remodulation permitted us to estimate the seasonal patterns of each series through time, and these are reported in figure 3 (b and c).¹⁰ The seasonal

⁹ Complex demodulates x_{jt} are defined $x_{jt} = L(x_t e^{i w_j t})$ where L is a low pass filter, x_t is the variable being studied at time t , and w_j is the frequency under investigation. The real and imaginary parts of the complex demodulate are calculated from $s_{jt}' = L(x_t \cos w_j t)$ and $s_{jt}'' = L(x_t \sin w_j t)$. The amplitude of oscillations at the frequency band centered at w_j at time t is given by $R_{jt} = 2[s_{jt}'^2 + s_{jt}''^2]^{1/2}$

and the phase is given by $\theta_{jt} = \tan^{-1} \frac{s_{jt}''}{s_{jt}'}$. It is possible

to remodulate the data, i.e., construct an estimated time series of the w_j frequency component, by using the relation $r_{jt} = 2s_{jt}' \cos w_j t + 2s_{jt}'' \sin w_j t$. If we are interested in the seasonal, we would demodulate at the twelve-month cycle and several of its harmonics, form the estimates of the r_{jt} at each of these frequencies, and construct an estimate of the seasonal from the relation $s_t = \sum_{j \in S} r_{jt}$ where S is

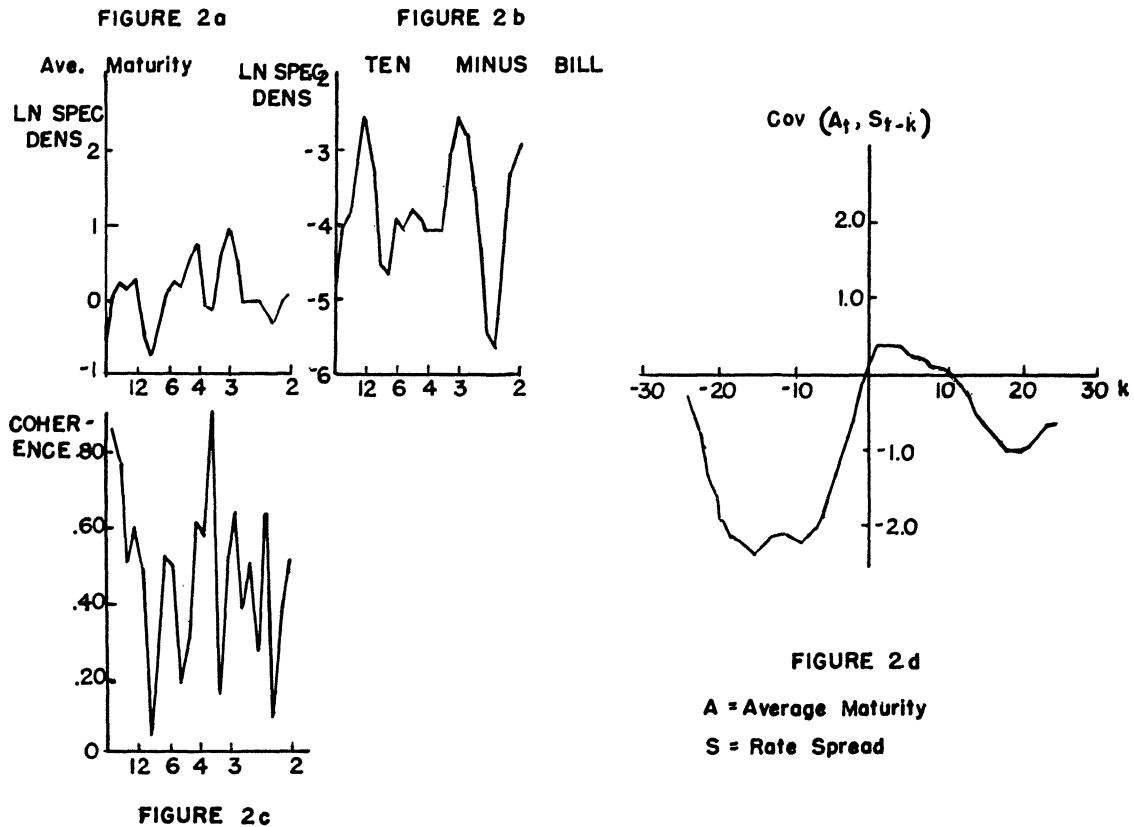
the set of frequencies corresponding to the seasonal. Our own procedure has been to demodulate at the twelve-month cycle and its first three harmonics. For the interest rates and their spread, we employed the filter which iterates twice

on $y_t = \frac{1}{2m+1} \sum_{i=-m}^{+m} x_{t-i}$. The parameter m was set at

eighteen. For the average maturity series, we iterated twice on a one-sided filter of length eighteen lags in order to derive seasonals for the early part of our period. See Godfrey [3] for a discussion of these techniques.

¹⁰ The estimated seasonal for the bill rate which is reported in figure 3a agrees fairly well with the pattern reported by Conard [1]. For the years 1955-1959, there is a seasonal trough in July followed by sharp increases in August and September. It is interesting to note that in each of these years, the Federal Reserve Bank of New York raised its discount rate in August or September, generally late August or early September. It is likely that the discount

FIGURE 2.



patterns, particularly for the first three years, appear to be similar, suggesting that the series were, at least over part of the period, directly related at the seasonal components. In addition, at least for the first several years, the two series appear to be close in phase. The figures in table 2, which records the coherence and phase of the cross-spectrum between differences in the rate spread and differences in the average maturity at various frequencies, permit us to study the timing more closely. The phase statistic recorded is the phase of the rate spread with respect to the average maturity. Notice that a minor correction of the phase statistic

rate changes were in response to the seasonal increases in the bill rate.

Selected Discount Rate Changes

Date	From	To
8/5/55	1 3/4	2
9/9/55	2	2 1/4
8/24/56	2 3/4	3
8/23/57	3	3 1/2
9/12/58	1 3/4	2
9/11/59	3 1/2	4

is necessitated by the difference in the times of the month at which the rate data and the average maturity data were collected. While the rate spread leads the maturity by a sizeable period at the longer periodicities, notice that the lead virtually vanishes at the shorter periodicities which correspond to the important peaks in the spectral density of the differenced maturity series. At these shorter periodicities the leads are clearly not significant at any usual significance level. This tends to confirm the closeness of phase suggested by the seasonal patterns obtained by remodulation. The positive association and the closeness of phase of the two series are at least consistent with a mechanism which requires that rate spreads adjust quickly in response to variations in the average maturity at these components of oscillation. However, it is equally clear that such debt-maturity effects on the yield spread were at best transitory since at the lower frequency components the spread displays a size-

FIGURE 3.

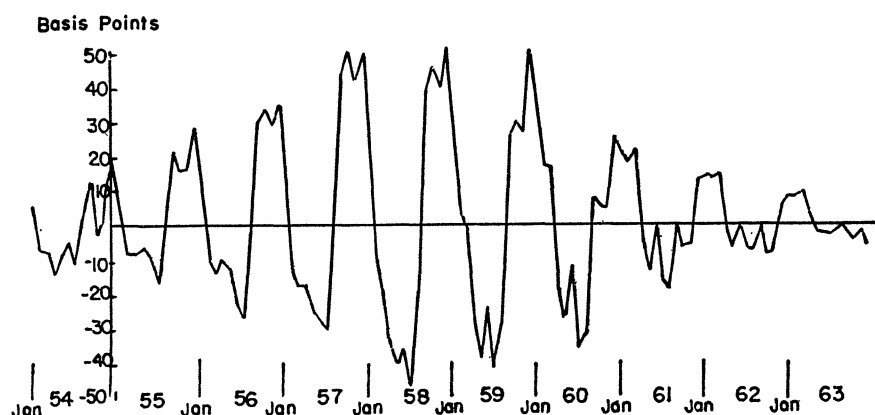


FIGURE 3 a

Bill Rate

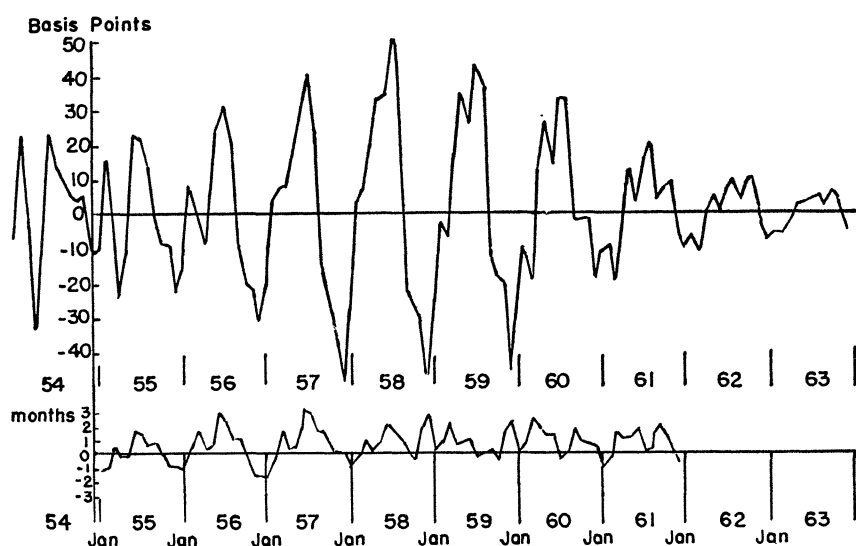


FIGURE 3 b

Ten Year-Bill

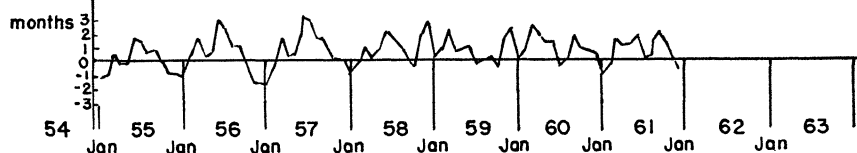


FIGURE 3 c

Average Maturity

TABLE 2.¹¹

Period	Coherence	Phase ^a	Phase ^a (corrected)	Corrected Phase Difference (months)
48	.86	-.71	-.78	-5.92
24	.79	-.50	-.63	-2.41
16	.51	-.46	-.66	-1.67
12	.61	-.22	-.48	-.92
6	.52	-.44	-.96	-.92
4	.61	.43	-.35	-.23
3	.51	.78	-.27	-.13
2	.52	.71	-.86	-.28

^a Recorded in radians.

¹¹ Since the average maturity series consists of end-of-month observations while the rate series consists of mid-month observations throughout most of the period, it was necessary to correct the phase statistic by adding one-half month to the lead of the spread over the maturity. This was accomplished by subtracting π/n from the calculated phase, where n is the length of period. To convert from radians to months, we simply multiplied the phase in radians by $n/2\pi$.

able lead over the maturity. Indeed, the cross-covariogram between the levels of the average maturity and the rate spread, which is recorded in figure 2d, shows clearly the impact of the lead of the rate spread over maturity at all of the lower frequency components. The highest covariances obtain at lags of the average maturity of one to five months behind the rate spread. The pattern of these timing relationships reveals that the direction of influence between the rate spread and the average maturity may well have gone two ways, as the considerations discussed in footnote 7 lead us to expect. At the longer period oscillations, the average maturity responded directly to the rate spread with a lag. This indicates that the government was adjusting the maturity structure in a fashion designed to minimize the interest cost of

TABLE 3. — FIRST DIFFERENCES (1951–1960)

Period	Coherence					
	3 mo Aaa	3 mo Baa	3 mo Comm. Paper	Govt. Long Aaa	Govt. Long Baa	Govt. Long Comm. Paper
60	.7552	.5810	.9764	.9381	.5258	.7245
30	.8202	.6750	.9429	.9612	.7553	.8482
20	.7708	.7078	.9331	.9809	.8033	.8451
15	.7499	.7062	.9467	.9548	.7030	.8970
12	.7390	.6639	.9614	.9237	.7516	.7302
Phase Shift ^a						
60	.3023	.9825	.1500	.2008	.5979	.1538
30	.0189	.6880	.2117	.1397	.6926	.4220
20	— .0463	.4145	.3308	.0535	.4343	.5397
15	— .1773	.3502	.2183	— .0227	.5414	.3252
12	— .6182	.0084	.1118	.0135	.6956	.5903
Gain ^b						
60	3.2955	2.2493	1.0050	1.0372	.6043	.2445
30	3.1997	3.0589	.9982	1.0459	.9771	.2859
20	2.8741	3.2819	1.0085	1.1028	1.1892	.3264
15	3.3936	3.4521	1.1227	.9886	.8992	.2821
12	3.2039	2.6650	1.1844	.8182	.6477	.2358

^a Recorded in radians.^b Gain of government yield with respect to corporate yield.

the debt. A rise in the rate spread, which generally occurred when long rates were low, tended to induce a lengthening of the maturity as the opportunity to issue longs at low rates of interest was taken. Similarly, a low rate spread induced a shortening of the maturity. While the timing of the relationship at the longer oscillations is consistent with this direction of influence, the timing at the higher frequencies we have studied indicates that here it is possibly the average maturity which produced the observed oscillations in the rate spread.¹²

III Relationships between Government and Corporate Rates

Table 3 records some results of a cross-spectral analysis of first differences in Aaa, Baa, and commercial paper rates against first differences in both treasury bill rates and the government long rate. Thirty was the maximal lag. The results are largely self-explanatory, and we shall confine ourselves to pointing out a few salient features of the results. First, while the coherences are generally very high, the Aaa is more closely related to the government rates than is the Baa index, and both are

more closely associated with the longer than with the shorter government yield. On the other hand, the short maturity commercial paper rate is more closely related to the bill rate than the long government rate. Second, notice that the government rates generally lead the corporate rates. The chief exception is the lead which the Aaa displays over the bill rate at frequencies close to the seasonal. Undoubtedly, this arises from the long-short spread phenomenon discussed above. Note also the considerable lag behind the Aaa which the Baa displays, since the Baa lags the governments by more at each frequency. The pattern which emerges is one in which the rates on higher quality instruments lead the rates on lower quality instruments at almost every frequency.¹³ Finally, notice that the amplitudes of the various instruments tend to vary inversely with maturity and quality.

¹³ The presence of this pattern at the cyclical frequencies is probably explained by the co-cyclical character of default rates. Near the upper turning point, default rates are relatively high and this tends to increase rates on the riskier instruments for some period. This effect is likely to continue even as the rates on the safer assets fall during the contraction phase of the cycle. Thus the gap between rates on safe and risky assets increases during this period. Of course, rates on the riskier instruments do fall later in the contraction, presumably with some lag behind the falling default rate. Clearly, this mechanism can also explain a lag of rates on risky assets behind rates on safe assets at the lower turning point.

¹² Conard [1] attributes the seasonal pattern in bill rates to the character of Treasury operations in the bill market over the year.

IV Conclusions

The statistics presented indicate that, despite wide differences in amplitude, a high level of association existed among interest rates on a variety of money and capital market instruments during the period examined. The closer the instruments were with respect to maturity and quality, the higher the association tended to be. The timing of the relationships at the various frequencies was such that the government long rates led all other rates, the lead generally increasing with shortening of the maturity and deterioration of quality. These findings about timing tend to support Okun's finding [8] that the long government rate was a key market rate over a period similar to the one studied here.

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